MACHINE LEARNING ASSIGNMENT 10

Name Harish Panpaliya

Branch and Division: CSAI-D

Roll no: 52

Assignment on Classification technique. Every year many students give the GRE exam to get admission in foreign Universities. The data set contains GRE Scores (out of 340), TOEFL Scores (out of 120), University Rating (out of 5), Statement of Purpose strength (out of 5), Letter of Recommendation strength (out of 5), Undergraduate GPA (out of 10), Research Experience (0=no, 1=yes), Admitted (0=no, 1=yes). Admitted is the target variable.

The counsellor of the firm is supposed check whether the student will get an admission or not based on his/her GRE score and Academic Score. So to help the counsellor to make appropriate decisions, build a machine learning model classifier using a Decision tree to predict whether a student will get admission or not. Apply Data pre-processing (Label Encoding, Data Transformation....) techniques if necessary.

1. Perform data-preparation (Train-Test Split)
2. Apply Machine Learning Algorithm
3. Evaluate Model.

Dataset: https://www.kaggle.com/mohansacharya/graduate-admissions

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the dataset
df = pd.read_csv('Admission_Predict_Ver1.1.csv')

# Display the first 5 rows
df.head()
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
0	1	337	118	4	4.5	4.5	9.65	1	0.92
1	2	324	107	4	4.0	4.5	8.87	1	0.76
2	3	316	104	3	3.0	3.5	8.00	1	0.72
3	4	322	110	3	3.5	2.5	8.67	1	0.80
4	5	314	103	2	2.0	3.0	8.21	0	0.65

Next steps: Generate code with df New interactive sheet

```
# 1. Clean column names (removing leading/trailing spaces)
df.columns = df.columns.str.strip()

# 2. Transform 'Chance of Admit' into a binary 'Admitted' variable
# We use a threshold of 0.7: 1 if Chance >= 0.7, else 0
df['Admitted'] = (df['Chance of Admit'] >= 0.7).astype(int)

# 3. Drop unnecessary columns
# 'Serial No.' is an identifier and 'Chance of Admit' is the old target
df_clean = df.drop(['Serial No.', 'Chance of Admit'], axis=1)

# Check the distribution of the new target variable
print(df_clean['Admitted'].value_counts())
df_clean.head()
```

```
Admitted
1    300
0    200
Name: count, dtype: int64
```

	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Admitted
0	337	118	4	4.5	4.5	9.65	1	1
1	324	107	4	4.0	4.5	8.87	1	1
2	316	104	3	3.0	3.5	8.00	1	1
3	322	110	3	3.5	2.5	8.67	1	1
4	314	103	2	2.0	3.0	8.21	0	0

Next steps: Generate code with df_clean New interactive sheet

Train-Test Split

```
# Define Features (X) and Target (y)
X = df_clean.drop('Admitted', axis=1)
y = df_clean['Admitted']

# Split data: 80% Training, 20% Testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Training set size: {X_train.shape}")
print(f"Testing set size: {X_test.shape}")
```

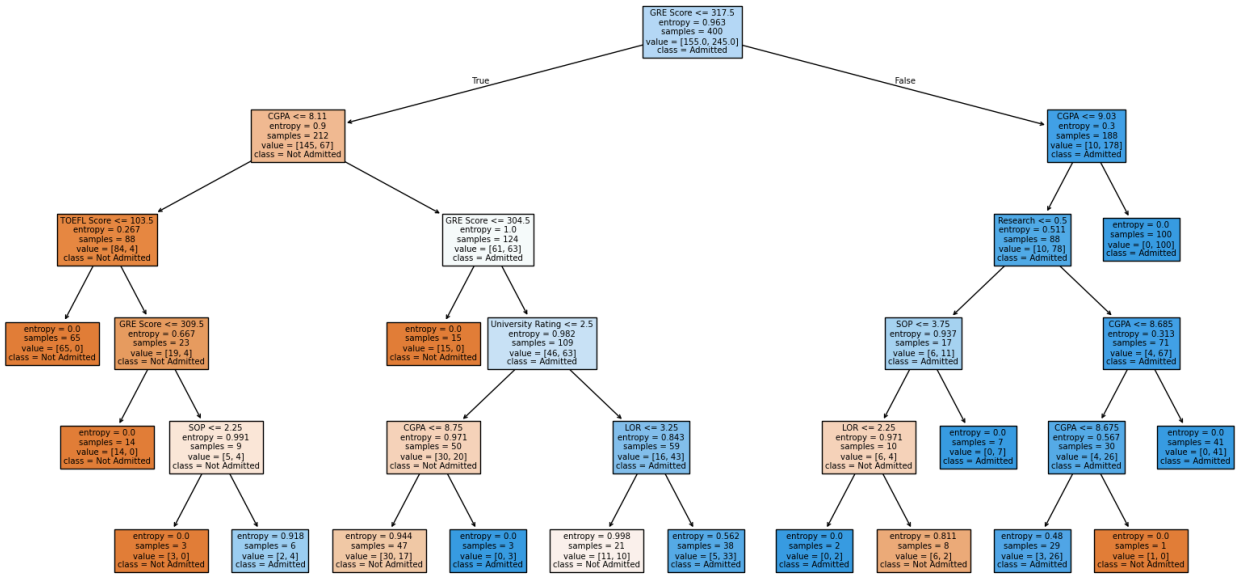
```
Training set size: (400, 7)
Testing set size: (100, 7)
```

Decision Tree

```
# Initialize the Classifier
dt_classifier = DecisionTreeClassifier(criterion='entropy', max_depth=5, random_state=42)

# Train the model
dt_classifier.fit(X_train, y_train)

# Optional: Visualize the tree structure
plt.figure(figsize=(20,10))
plot_tree(dt_classifier, feature_names=X.columns, class_names=['Not Admitted', 'Admitted'], filled=True)
plt.show()
```



Model Evaluation

```
# Make predictions
y_pred = dt_classifier.predict(X_test)

# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Overall Accuracy: {accuracy * 100:.2f}%")

# Confusion Matrix
print("\nConfusion Matrix:")
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()

# Detailed Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Overall Accuracy: 82.00%
```